二次域的同构

Lemma：R和R'是环，$f: R \longrightarrow R'$是环同构，则有：$f^{-1}: R' \longrightarrow R$是环同构，且有：$f^{-1}f = id_R$ 且 $ff^{-1} = id_{R'}$

Proof：$\because f: R \longrightarrow R'$是环同构　$\therefore f$是双射　$\therefore f$是可逆映射

$\therefore$存在映射$f^{-1}: R' \longrightarrow R$，使得 $f^{-1}f = id_R$ 且 $ff^{-1} = id_{R'}$

$\because f: R \rightarrow R'$是可逆映射　$\therefore f^{-1}: R' \longrightarrow R$也是可逆映射

$\therefore f^{-1}: R' \longrightarrow R$是双射.

$\because f$是环同构　$\therefore f^{-1}: R' \longrightarrow R$是环同态

$\therefore f^{-1}: R' \longrightarrow R$是环同构，且有 $f^{-1}f = id_R$ 且 $ff^{-1} = id_{R'}$　$\square$

Lemma：设D_1和D_2为无平方因子的非零整数，$D_1, D_2 \neq 1$.

$f: \mathbb{Q}(\sqrt{D_1}) \xrightarrow{\sim} \mathbb{Q}(\sqrt{D_2})$是域的同构，则有：

① 对$\forall x \in \mathbb{Q}$，有：$f(x) = x$

② 对$\forall y \in \mathbb{Q}$，有：$f^{-1}(y) = y$

Proof：$\because \mathbb{Q}(\sqrt{D_1})$和$\mathbb{Q}(\sqrt{D_2})$是域　$\therefore \mathbb{Q}(\sqrt{D_1})$和$\mathbb{Q}(\sqrt{D_2})$是环.

$\because f: \mathbb{Q}(\sqrt{D_1}) \xrightarrow{\sim} \mathbb{Q}(\sqrt{D_2})$是域的同构

$\therefore f: \mathbb{Q}(\sqrt{D_1}) \xrightarrow{\sim} \mathbb{Q}(\sqrt{D_2})$是环的同构

$\therefore f(0) = 0$，$f(1) = 1$.

对$\forall x\in\mathbb{Q}$，有：$x=x+0\sqrt{D_1}\in\mathbb{Q}(\sqrt{D_1})$，$x=x+0\sqrt{D_2}\in\mathbb{Q}(\sqrt{D_2})$

$\therefore\ \mathbb{Q}\subseteq\mathbb{Q}(\sqrt{D_1})$，$\mathbb{Q}\subseteq\mathbb{Q}(\sqrt{D_2})$

对$\forall n\in\mathbb{Z}$，有：

(i). 若$n>0$，则$n\in\mathbb{Z}_{\geqslant 1}$ $\quad\therefore\ n=\underbrace{1+\cdots+1}_{n个1}$

$\therefore\ f(n)=f(\underbrace{1+\cdots+1}_{n个1})=\underbrace{f(1)+\cdots+f(1)}_{n个f(1)}=\underbrace{1+\cdots+1}_{n个1}=n$

(ii) 若$n=0$，则有：$f(n)=f(0)=0=n$

(iii) 若$n<0$，则有：$-n>0$ $\quad\therefore -n\in\mathbb{Z}_{\geqslant 1}$ $\quad\therefore\ -n=\underbrace{1+\cdots+1}_{(-n)个1}$

$\therefore\ f(n)=f(-(-n))=-f(-n)=-(-n)=n$

$\therefore$ 对$\forall n\in\mathbb{Z}$，有：$f(n)=n$

对$\forall x\in\mathbb{Q}$，有：$\exists\, p,q\in\mathbb{Z}$，$q\neq 0$，$\gcd(p,q)=1$，s.t. $x=\frac{p}{q}=pq^{-1}$

$\therefore\ f(x)=f(pq^{-1})=f(p)f(q^{-1})=f(p)\cdot(f(q))^{-1}=p\cdot q^{-1}=x$. ①得证.

$\because\ f:\mathbb{Q}(\sqrt{D_1})\xrightarrow{\sim}\mathbb{Q}(\sqrt{D_2})$是环同构 $\quad\therefore\ f^{-1}:\mathbb{Q}(\sqrt{D_2})\xrightarrow{\sim}\mathbb{Q}(\sqrt{D_1})$是环同构.

对$\forall y\in\mathbb{Q}$，有：$y\in\mathbb{Q}(\sqrt{D_1})$，~~ ~~ $y\in\mathbb{Q}(\sqrt{D_2})$，$f^{-1}(y)\in\mathbb{Q}(\sqrt{D_1})$

$\because\ f:\mathbb{Q}(\sqrt{D_1})\longrightarrow\mathbb{Q}(\sqrt{D_2})$是环同构 $\quad\therefore f$是双射. $\quad\therefore f$是单射.

$\because\ f^{-1}(y)\in\mathbb{Q}(\sqrt{D_1})$

$\therefore\ f(f^{-1}(y))=(f\circ f^{-1})(y)=id_{\mathbb{Q}(\sqrt{D_2})}(y)=y=f(y)$

$\therefore\ f^{-1}(y)=y$. ②得证. $\square$

Lemma：设 D_1 和 D_2 为无平方因子的非零整数，$D_1, D_2 \neq 1$，则有：

$$\mathbb{Q}(\sqrt{D_1}) \simeq \mathbb{Q}(\sqrt{D_2}) \iff D_1 = D_2$$

Proof：($\Leftarrow$)：$\because D_1 = D_2 \quad \therefore \sqrt{D_1} = \sqrt{D_2} \quad \therefore \mathbb{Q}(\sqrt{D_1}) = \mathbb{Q}(\sqrt{D_2})$

$\therefore \mathbb{Q}(\sqrt{D_1}) \simeq \mathbb{Q}(\sqrt{D_2})$

($\Rightarrow$)：$\because \mathbb{Q}(\sqrt{D_1}) \simeq \mathbb{Q}(\sqrt{D_2})$ $\quad\therefore$ 设 $f: \mathbb{Q}(\sqrt{D_1}) \xrightarrow{\sim} \mathbb{Q}(\sqrt{D_2})$ 为域的同构.

$\because \sqrt{D_1} = 0 + 1\sqrt{D_1} \in \mathbb{Q}(\sqrt{D_1}) \quad \therefore f(\sqrt{D_1}) \in \mathbb{Q}(\sqrt{D_2})$

$\therefore \exists\, c, d \in \mathbb{Q}$，s.t. $f(\sqrt{D_1}) = c + d\sqrt{D_2}$

$\because (\sqrt{D_1})^2 = D_1 = D_1 + 0\sqrt{D_1} \in \mathbb{Q}(\sqrt{D_1})$

$\therefore f((\sqrt{D_1})^2) = f(D_1)$

$\therefore f(D_1) = f((\sqrt{D_1})^2) = f(\sqrt{D_1} \cdot \sqrt{D_1}) = f(\sqrt{D_1}) \cdot f(\sqrt{D_1}) = (c + d\sqrt{D_2}) \cdot (c + d\sqrt{D_2})$

$$= (c + d\sqrt{D_2})^2 = c^2 + d^2 D_2 + 2cd\sqrt{D_2} \in \mathbb{Q}(\sqrt{D_2})$$

$\because D_1 \in \mathbb{Q} \quad \therefore f(D_1) = D_1 \quad \therefore D_1 = c^2 + d^2 D_2 + 2cd\sqrt{D_2} \in \mathbb{Q}(\sqrt{D_2})$

$\therefore D_1 + 0\sqrt{D_2} = (c^2 + d^2 D_2) + 2cd\sqrt{D_2} \in \mathbb{Q}(\sqrt{D_2})$

$\therefore D_1 = c^2 + d^2 D_2$ 且 $2cd = 0 \quad \because 2cd = 0 \quad \therefore cd = 0 \quad \therefore c = 0$ 或 $d = 0$

若 $d = 0$，则有：$D_1 = c^2 + d^2 D_2 = c^2 + 0 \cdot D_2 = c^2$

$\therefore \sqrt{D_1} = \sqrt{c^2} = |c| \in \mathbb{Q}$.

当 $D_1 > 0$ 时，D_1 是无平方因子的非零正整数，$D_1 \neq 1 \quad \therefore \sqrt{D_1} \in \mathbb{R} \setminus \mathbb{Q}$. 矛盾.

当 $D_1 < 0$ 时，$\sqrt{D_1} = \sqrt{-(-D_1)} = \sqrt{(-1)(-D_1)} = \sqrt{-1} \cdot \sqrt{-D_1} = \sqrt{-D_1}\, i \notin \mathbb{Q}$. 矛盾.

$\therefore d \neq 0 \quad \therefore c = 0 \quad \therefore D_1 = d^2 D_2$

$\because \alpha \in \mathbb{Q}$ $\therefore \exists p, q \in \mathbb{Z}$, $q \neq 0$, $\gcd(p,q)=1$, s.t. $\alpha = \frac{p}{q}$

$\therefore D_1 = \alpha^2 D_2 = \left(\frac{p}{q}\right)^2 D_2 = \frac{p^2}{q^2} D_2$ $\because q \neq 0$ $\therefore q^2 \neq 0$ $\therefore q^2 D_1 = p^2 D_2$

$\because \frac{p^2 D_2}{q^2} = D_1 \in \mathbb{Z}$ $\therefore q^2 \mid p^2 D_2$

$\because \gcd(q^2, p^2) = \gcd(p^2, q^2) = (\gcd(p,q))^2 = 1^2 = 1$

$\because q^2 \in \mathbb{Z}$, $p^2 \in \mathbb{Z}$, $D_2 \in \mathbb{Z}$, $q^2 \neq 0$, $\gcd(q^2, p^2) = 1$, $q^2 \mid p^2 D_2$

$\therefore q^2 \mid D_2$ ~~D_2有平~~ 假设 $q \neq \pm 1$，则有：$q \neq 0$ 且 $q \neq \pm 1$

$\therefore q \in \mathbb{Z}_{\geqslant 2}$ 或 $q \in \mathbb{Z}_{\leqslant -2}$ $\therefore D_2$有平方因子 矛盾. $\therefore q = \pm 1$ $\therefore q^2 = 1$

$\therefore D_1 = p^2 D_2$.

假设 $p=0$，则有：$D_1 = 0^2 D_2 = 0$. 矛盾. $\therefore p \neq 0$. $\therefore p^2 \neq 0$

$\because \frac{D_1}{p^2} = D_2 \in \mathbb{Z}$ $\therefore p^2 \mid D_1$.

假设 $p \neq \pm 1$，则有：$p \neq 0$ 且 $p \neq \pm 1$ $\therefore p \in \mathbb{Z}_{\geqslant 2}$ 或 $p \in \mathbb{Z}_{\leqslant -2}$

$\therefore D_1$有平方因子. 矛盾. $\therefore p = \pm 1$ $\therefore p^2 = 1$ $\therefore D_1 = D_2$ $\square$